Fakultät für Informatik und Mathematik
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Winter semester 2026/2027

Exercises for Lecture: Virtual Machines and Runtime Systems

Exercise Sheet 4 (Prolog, Logic Programming in Java)

Problem 1 (Numbers and Lists in Prolog)

Using compound terms one can represent more complex structures than only constants in Prolog. For example, the natural numbers can be represented as

`z`, `s(z)`, `s(s(z))`, ...

where `z` is a constant and the unary constructor `s` makes a number one bigger from a term representing an existing number. Similarly, one can represent lists using the constant `nil` and a binary constructor `cons`, e.g.,

`nil`, `cons(z,nil)`, `cons(z,cons(s(z),nil))`, `cons(z,cons(s(z),cons(s(s(z)),nil)))`, ...

for the lists `[]`, `[0]`, `[0,1]`, `[0,1,2]`. Compound terms can contain variables, too; for example, one can use `s(X)` or `cons(s(X),Y)` in clauses and queries. Addition on natural numbers can be defined as follows:

```
add(z,B,B).                          % z+B=B
add(s(A),B,s(C)) :- add(A,B,C)       % A+B=C  =>  s(A)+B=s(C)
```

(a) Define a predicate `leq(A,B)` that is satisfied for two natural numbers `A` and `B` if and only if `A` is less than or equal to `B`.

(b) Define a predicate `allleq(A,L)` that holds for a natural number `A` and a list of natural numbers `L` if and only if `A` is less than or equal to all the elements in the list `L`. For example, `allleq(s(z), cons(s(z),cons(s(s(z)),nil)))` holds.

(c) Define a predicate `sorted(L)` that holds for a list of natural numbers `L` if and only if `L` is sorted in ascending order. Test your implementation with the following queries:

```
sorted(cons(z,cons(s(z),cons(s(s(z)),nil)))).
sorted(cons(z,cons(X,cons(s(s(z)),nil)))).
```

(d) (Discussed in next week's tutorial when the lecture discussed how to implement Prolog in Java:) Implement the tasks (a)-(c) in Java by extending the program `prolog.zip` (see Stud.IP).

Note:

- Lists can also be written in Prolog using the syntax `[1,2,3]` etc. The syntax `cons(A,L)` (for a list `L`) corresponds to `[A|L]`.
- In the CIP pool, SWI-Prolog is installed as `/home/groessli/bin/swipl`. Use the command `[prg].` to load a file `prg.pl` into the interpreter.

You can hand-in your programs by uploading them in the folder "Submissions" in Stud.IP (hand-in is optional).

Due date: Thursday, November 12 at 18:00, discussion on November 13, 2026